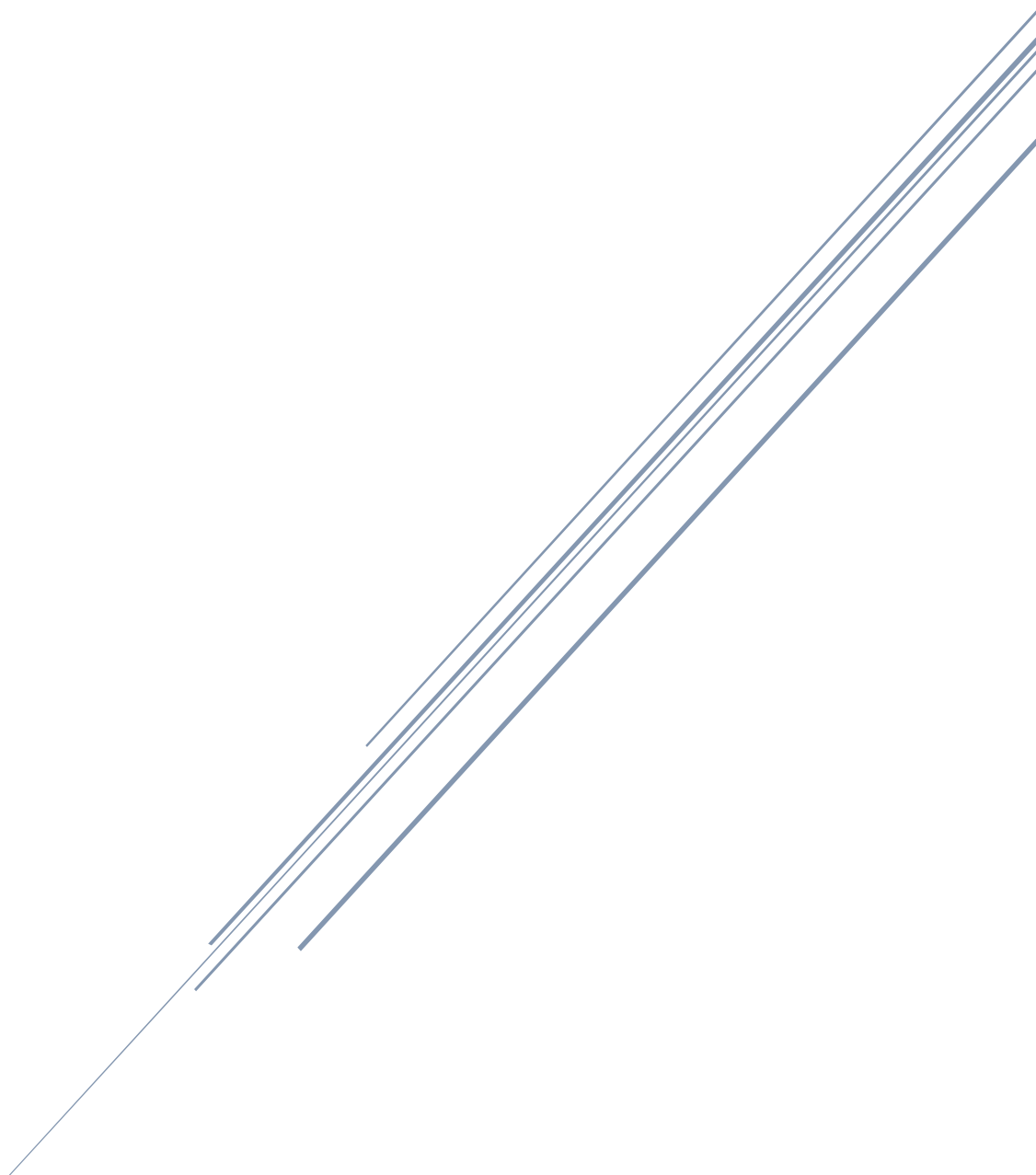


JSM82MS

ESP32 calculator



Mongolce

Table of Contents

Entering Special Mode	2
Main Menu Navigation	2
Opening an application	2
WiFi ON/OFF	2
Navigating Notes and Formulas	2
Unit Converter.....	2
Numeral / Programmer Calculator	2
How to Enter Numbers and Bases	3
Games	3
Snake	4
XOX (Tic-Tac-Toe).....	4
Pong	4
AI Assistant.....	4
Typing on the Keyboard	4
Sending and Reading Responses.....	4
Web Data Editor.....	5
How to Connect	5

Entering Special Mode

To enter special mode perform this operation: **ALPHA****MODE**

And to go back press **ON**

Also calculator has a safe screen that can be entered using the **Ans** key.

Main Menu Navigation

When you enter Special Mode, you will see a menu with numbered items (1-8), WiFi status (9), and your battery percentage.

Opening an application: Press the number button (1 through 8) on the keypad that corresponds to the application you want to open. You do not use **▲** or **▼** here.

WiFi ON/OFF: Press the 9 button to toggle the WiFi connection on or off.

Navigating Notes and Formulas

Once you enter the Notes or Formulas application:

- Press the **▲** and **▼** buttons to scroll through the list of available notes or formulas.
- And the **=** (Equals) button to open the selected note or formula.
- While reading a note or formula, press the **DEL** button to return to the list.
- From the list view, press **AC** to exit completely and return to the Main Menu.

Unit Converter

The converter guides you through a step-by-step process:

1. Select Category: Use **▲**/**▼** to select a category (e.g., Length, Weight), then press **=** to confirm.
2. Select "From" Unit: Use **▲**/**▼** to select the starting unit, then press **=**.
3. Select "To" Unit: Use **▲**/**▼** to select the target unit, then press **=**.
4. Enter Value: Use the number keypad to enter the value you want to convert.
 - Press **=** to calculate the result.
 - Press **DEL** to backspace your input.

At any step, press **DEL** to go back one step, or press **AC** to step back and eventually return to the Main Menu.

Numeral / Programmer Calculator

The Numeral tool allows you to perform calculations across different number bases (Decimal, Binary, Octal, and Hexadecimal).

How to Enter Numbers and Bases

You must specify the base before typing the number.

1. Type the base of your number (e.g., 2 for Binary, 10 for Decimal, 16 for Hexadecimal).
2. Press the  (Comma) button to confirm the base.
3. Now, type your number.

Typing Hexadecimal Letters (A-F): If you are working in Base 16, use the following specific buttons to type letters:

- A =  button
- B =  button
- C =  button
- D =  button
- E =  button
- F =  button

Mathematical Operations: Press an operator (, , , ). The calculator will then wait for you to specify the base for the *next* number. Once your equation is ready, press  to evaluate it.

Changing the Output Base: Once you have your result, press the  button to cycle the displayed result between different bases (Base 2, 8, 16, and 10).

Engineering / Remainder Mode: If you are doing division and want to see the remainder, press the  button. The screen will display the whole number result followed by an "R" and the remainder (for example: = 5 R 2). Press  again to toggle this off.

Clearing and Exiting:

- Press  to erase the last typed character.
- Press  to clear the current equation and exit back to the Main Menu.

Games

When you open the Games menu, you will see a selection screen.

- Press  or  to move the cursor between Snake, XOX, and Pong.
- Press  (Equals) to start the selected game.
- Press  to return to the Main Menu.

Snake

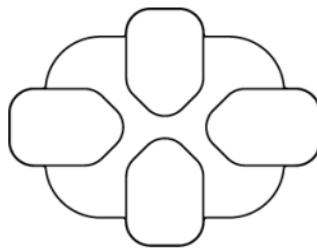
- Use the , ,  and  buttons to change the snake's direction.
- Collect the red food squares to grow your snake and increase your score.
- If you hit the walls or yourself, the game ends and returns to the Game Selection menu. You can also press  at any time to exit.

XOX (Tic-Tac-Toe)

- Use , ,  and  to move the selection square around the 3x3 grid.
- Press  (Equals) to place your "X" or "O" in the selected box.
- The game automatically returns to the Game Selection menu when someone wins or if there is a draw. You can also press  to exit early.

Pong

- **Player 1 (Left Paddle):** Press **4** to move up, and **1** to move down.
- **Player 2 (Right Paddle):** Press **6** to move up, and **3** to move down.
- Press  to clear the scores and return to the Game Selection menu.



AI Assistant

This tool allows you to write prompts using an on-screen keyboard and send them to the Gemini AI.

Typing on the Keyboard

- Use the directional buttons (, , , ) OR the number pad (8 for up, 5 for down, 4 for left, 6 for right) to move the highlight across the letters.
- Press the  (Comma) button to type the highlighted character into your prompt.
- Press  to delete the last character you typed.
- **Switch Keyboard Modes:**
 - Press the  button to toggle between lowercase and UPPERCASE (Caps) letters.
 - Press the  button to switch to numbers and symbols.

Sending and Reading Responses

- Once your prompt is ready, press  (Equals). The screen will say "Thinking..." while it fetches the response over WiFi.

- When the response appears, use  (or **8**) and  (or **5**) to scroll through the text line by line.
- Press  to close the response screen and return to the typing keyboard.
- Press  to completely exit the AI Assistant and return to the Main Menu.

Web Data Editor

This mode turns your calculator into a Wi-Fi Access Point so you can edit its data from another device. When you select this menu, the calculator will disconnect from its current network and create its own Wi-Fi network named "**Calculator**".

How to Connect

- Connect your smartphone or computer to the Wi-Fi network called "**Calculator**".
 - Open a web browser on that device and go to the address: **192.168.4.1/edit**
- Press . This will stop the server, turn off the Wi-Fi Access Point, and return you to the Main Menu.